

(a) reg→reg fmax [MHz]

(b) worst hold slack [ps]

Keccak, one round per clock, 20 ns	99	181	272	+723	+216	+37
Keccak, row-serialized, 20 ns	71	130	202	+717	+212	+31
original NTT, dual-port store, 20 ns	23	47	77	+512	+128	-11
iteration 1: 12-bit store, 20 ns	25	52	85	+507	+125	-14
iteration 1: 12-bit store + pipeline, 12 ns	43	89	145	+538	+144	-2
iteration 1: 12-bit store + pipeline, 20 ns	40	83	134	+525	+136	-8
original NTT, single-port store, 20 ns	24	50	80	+521	+134	-8
	slow 100 °C, 1.60 V	typical 25 °C, 1.80 V	fast −40 °C, 1.95 V	slow 100 °C, 1.60 V	typical 25 °C, 1.80 V	fast −40 °C, 1.95 V

Routed blocks at their own targets, OpenSTA with extracted parasitics; red: hold violated (before the hold margin)